

Antennas



Lebanese University
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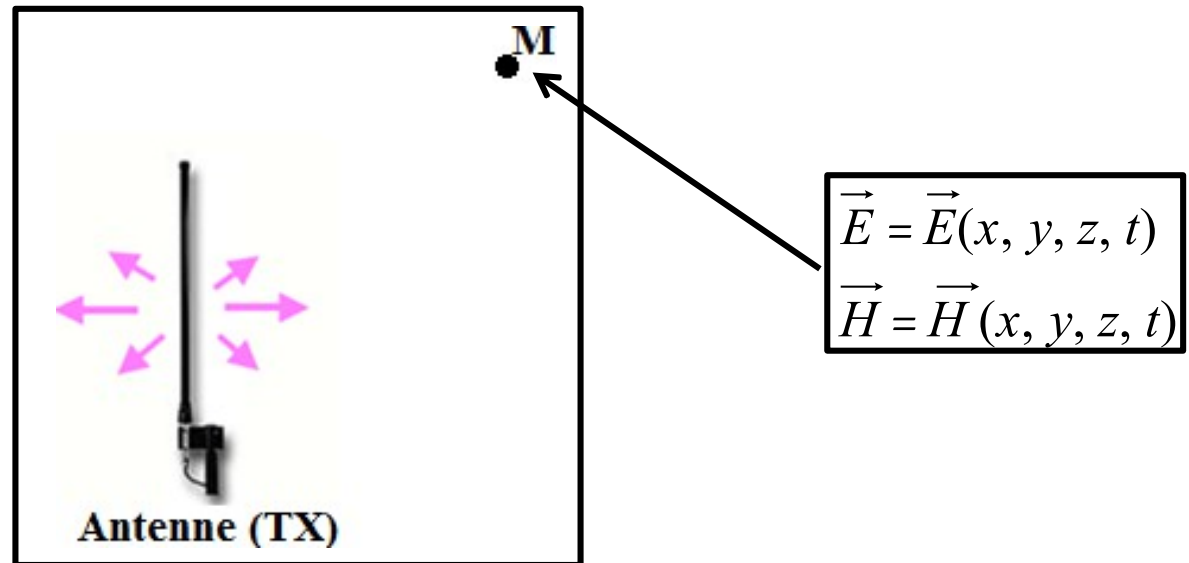
2023-2024

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Radiation electromagnetism

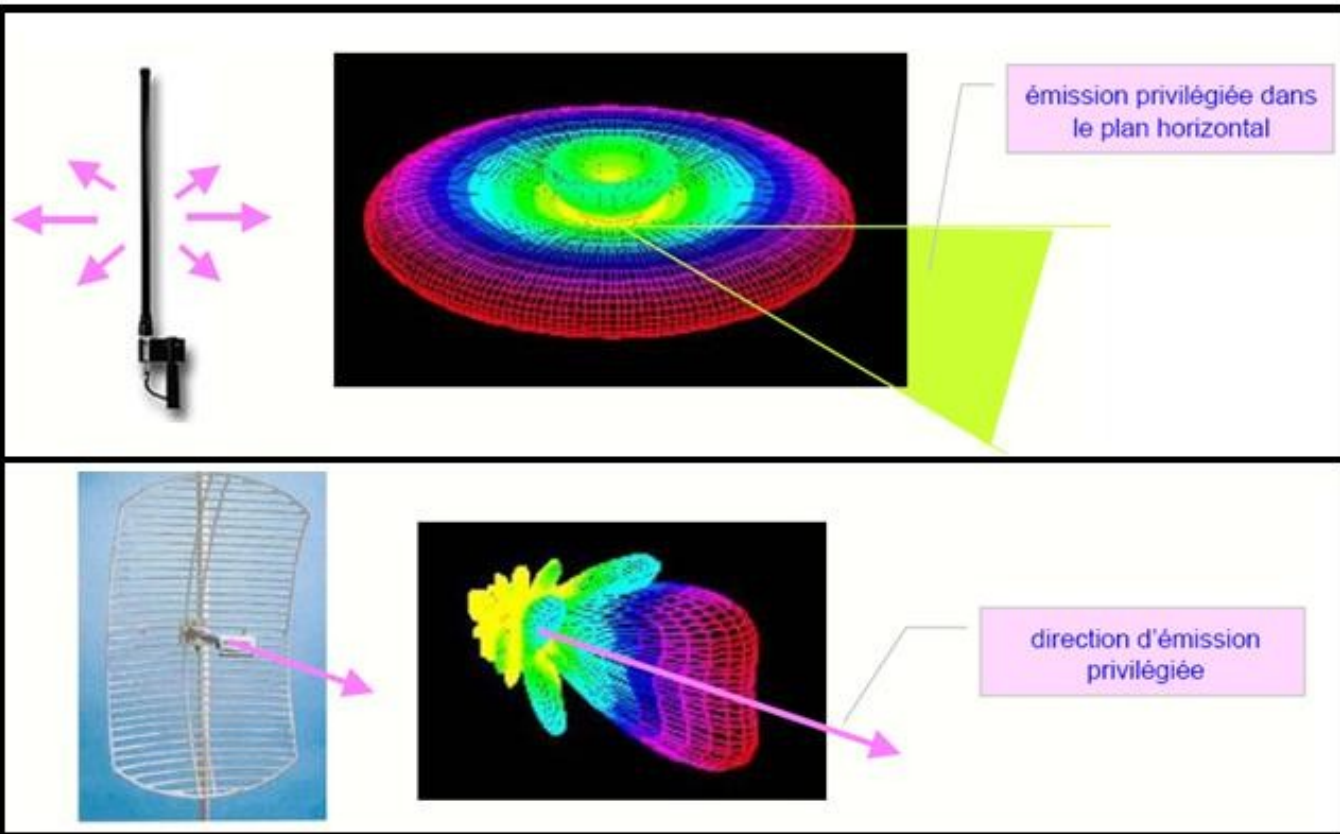
- Maxwell's equations $\forall \forall$
- Electromagnetic wave (OEM) = electric field (E) + magnetic field (H)
- Radiation: we'll calculate the OEM (E and H) at a point M in space around the antenna, whatever t is.



Antenna parameters

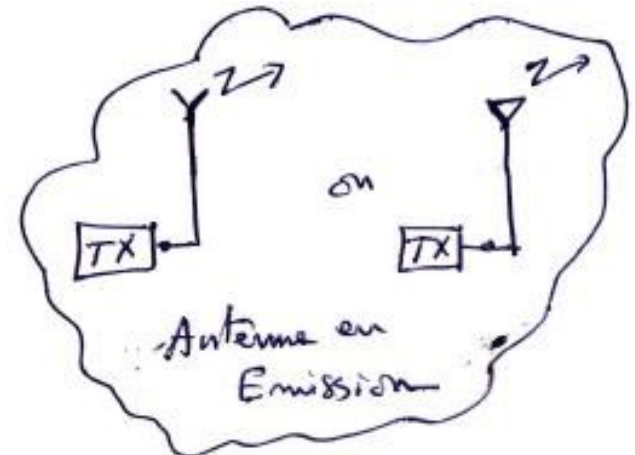
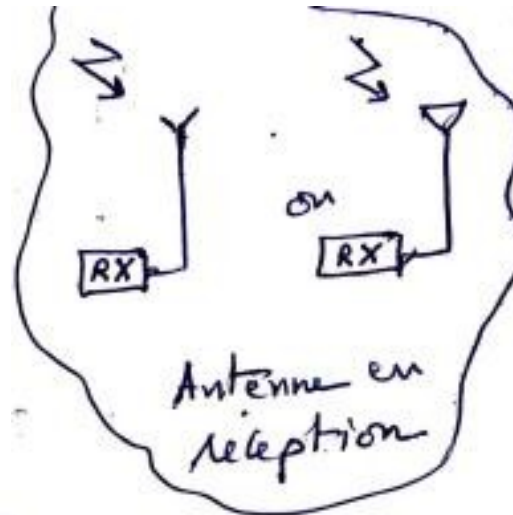
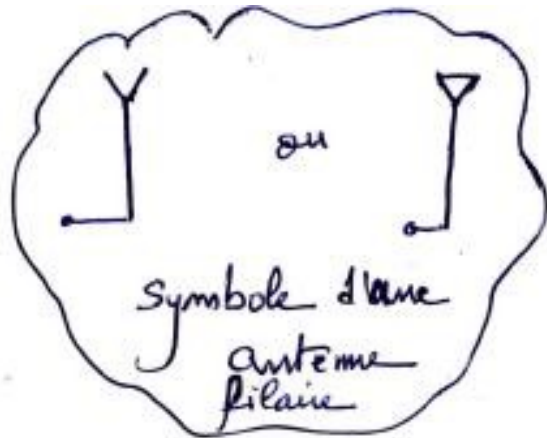
Radiation resistance, gain, directivity, bandwidth, radiation diagram,

Examples of
radiation
diagrams



Elementary and wire antennas

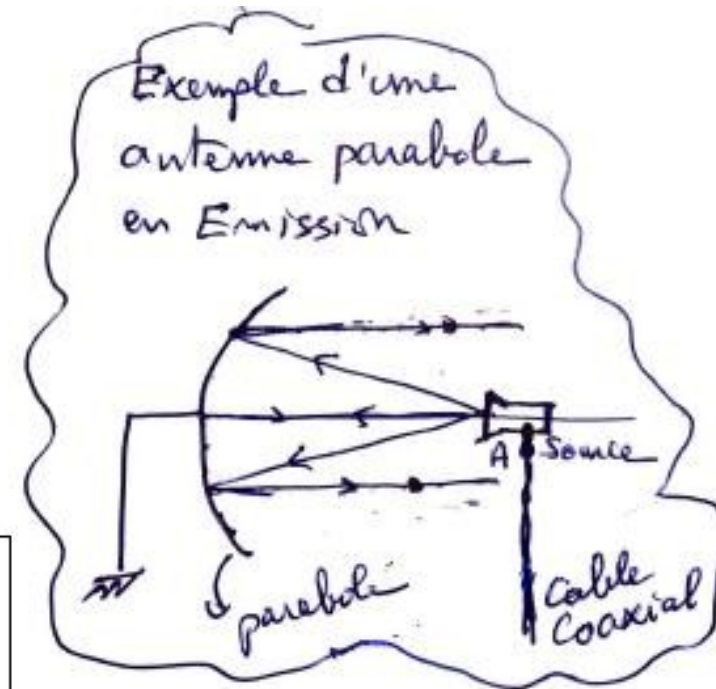
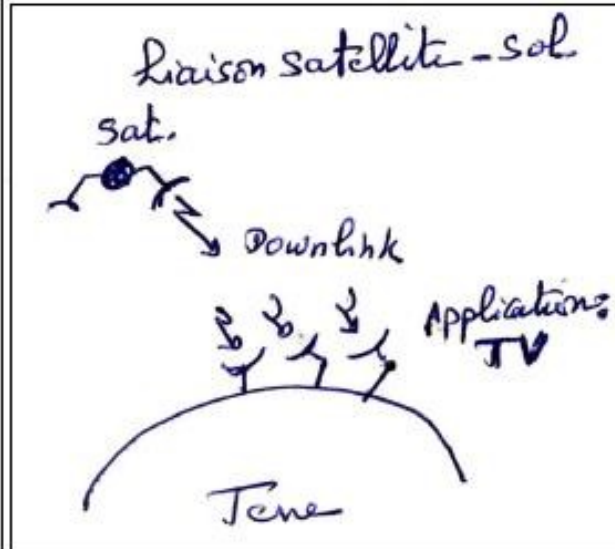
- The elementary antenna is a point antenna.
- Wired antenna :



Aperture antennas



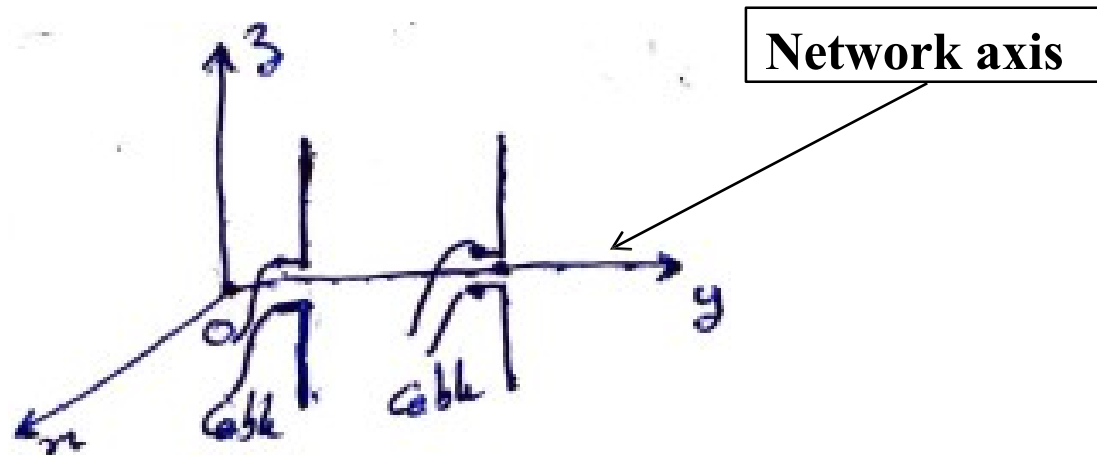
utilisation d'une antenne
parabole :
Liaison Sol-Satellite



Array antennas or antenna arrays

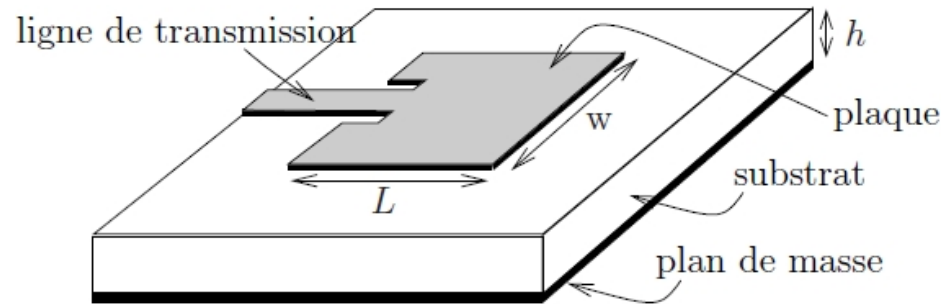
An antenna array is a set of antennas or elements.

**Ex: Uniform two-element linear array where
with two antennas (the two elements are identical).**

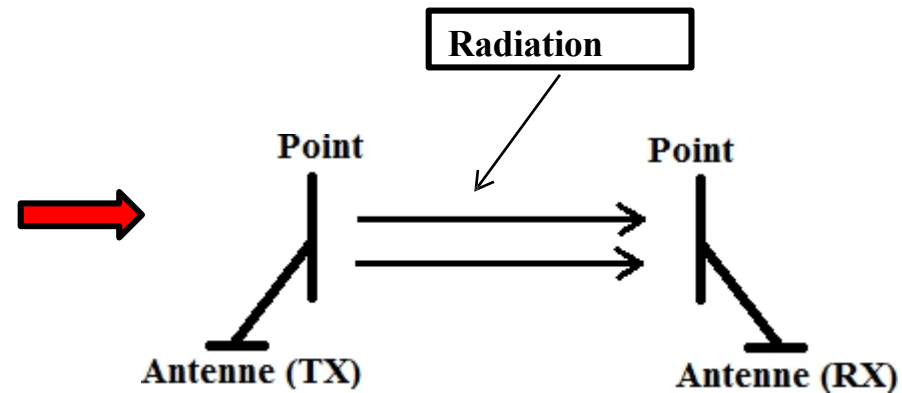


Planar or patch antenna

Ex.:

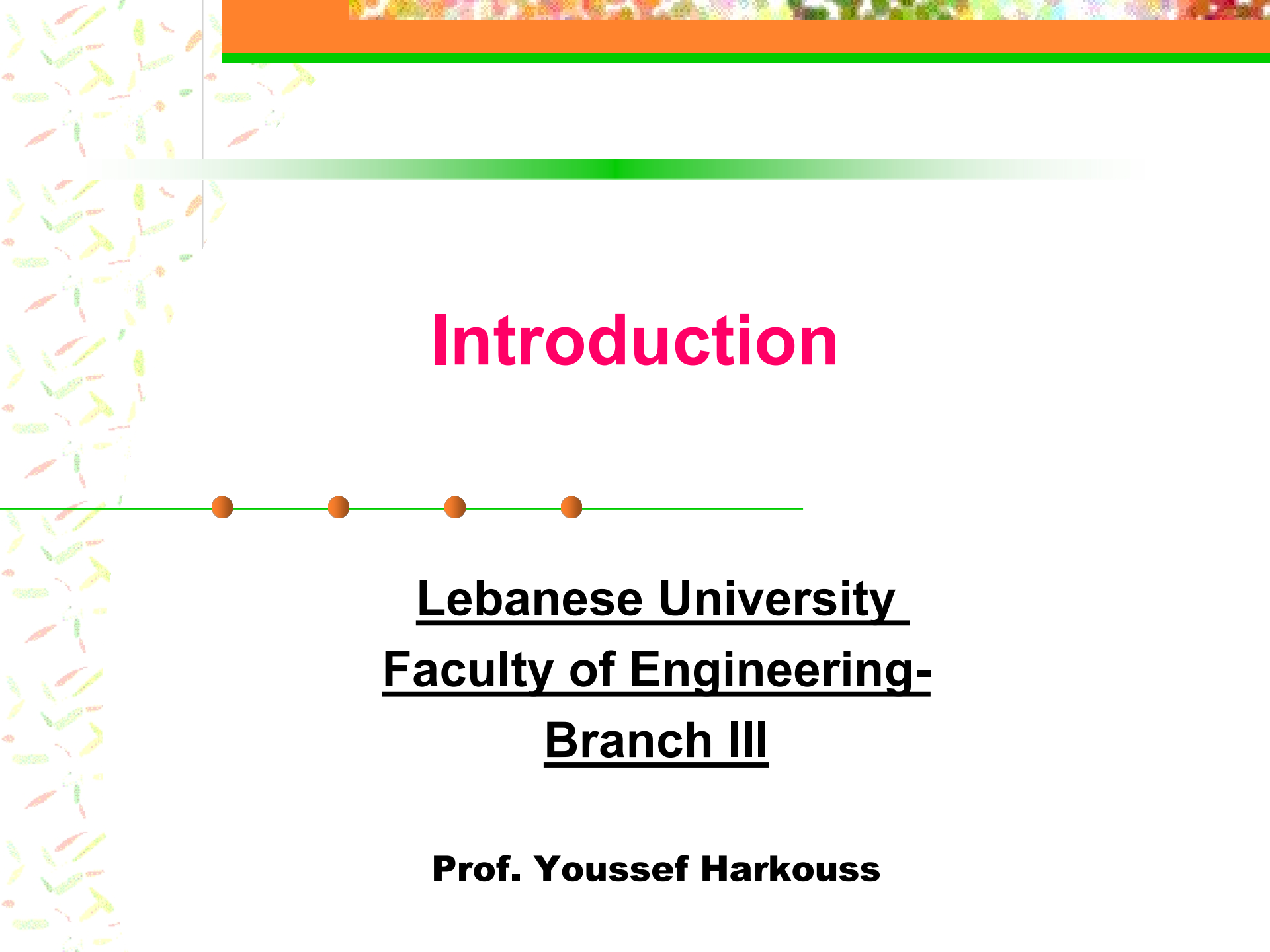


Antenna wifi panel-patch: a directional antenna. It ensures a connection point à point, n° well at inside also outdoor (supplier: TP-LINK).



References

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- Thomas A. Milligan, **Modern antenna design**, 2nd edition, Wiley-IEEE Press, 2005.
- C. A. Balanis, **Antenna theory: Analysis and design**, Wiley-interscience, 3rd edition, 2005.
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Introduction

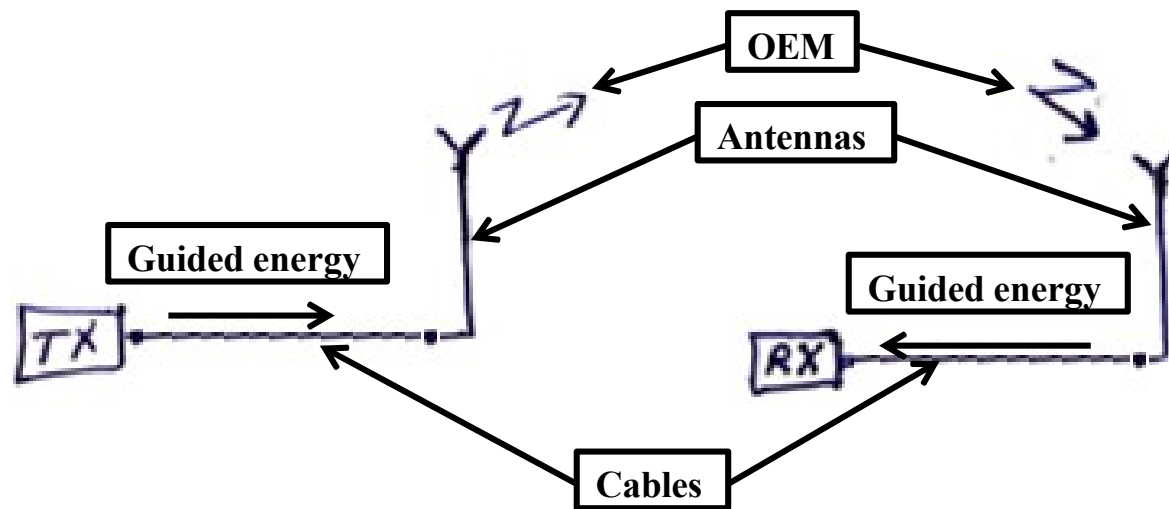
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What are we going to talk about?

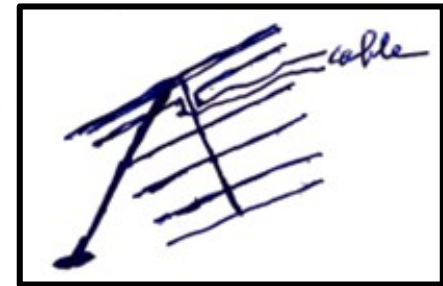
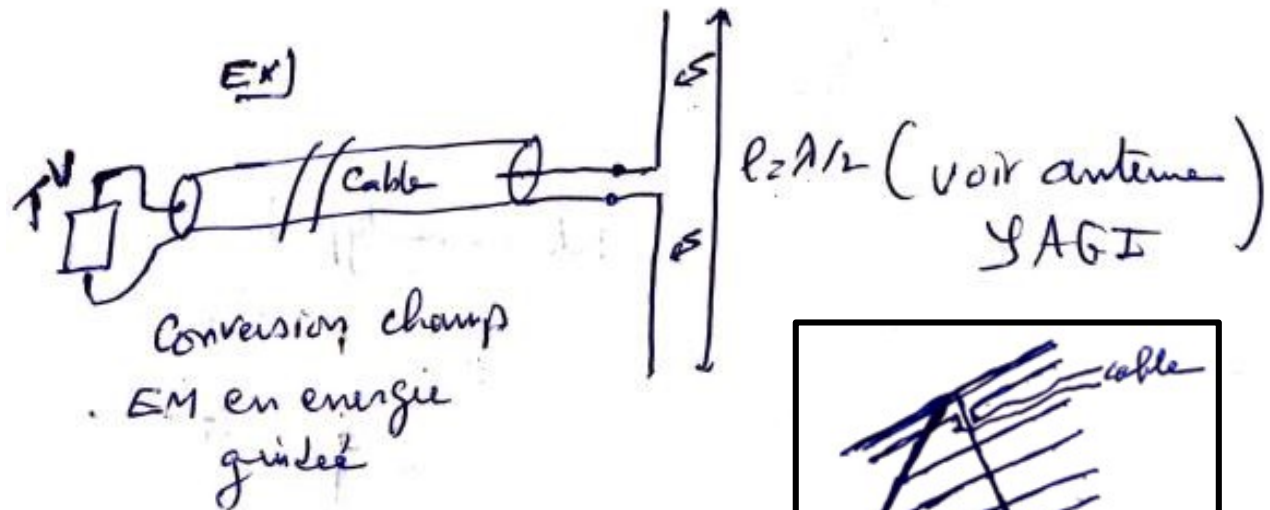
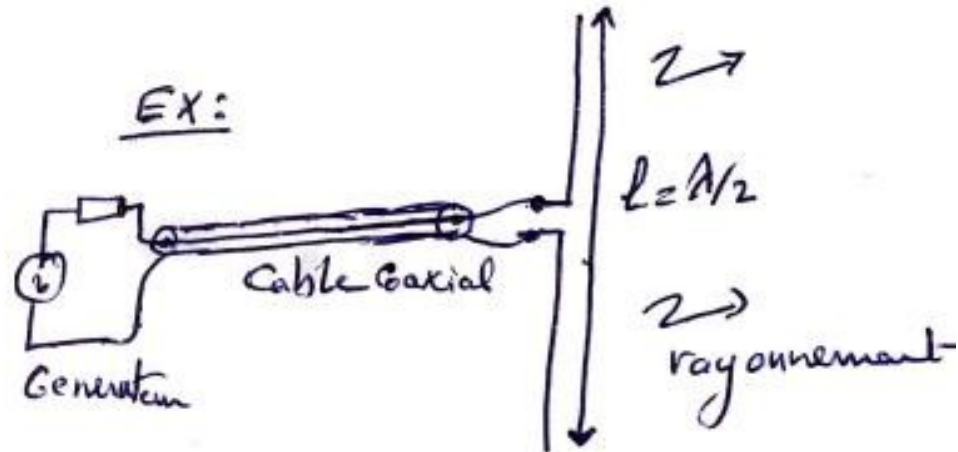
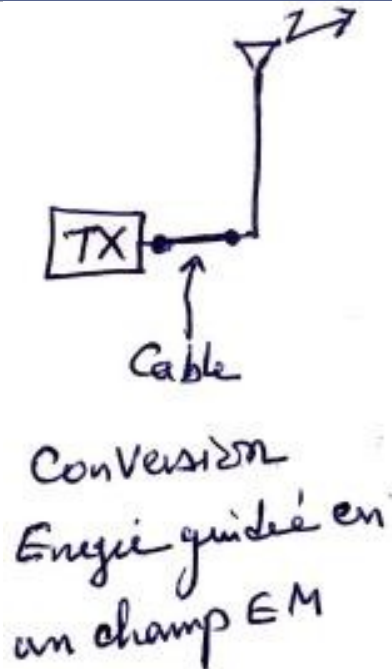
- We're going to talk about antennas (or aerials), devices that transpose guided energy into an electromagnetic field (or OEM electromagnetic wave) radiated into free space (or vice versa).



- There are many different types of antenna:
 - in a wide variety of shapes and sizes,
 - more or less complex operating modes.

We will define the characteristics and properties common to these different devices, as well as their areas of application.

Guided energy - EM field conversion



Antennas in everyday life

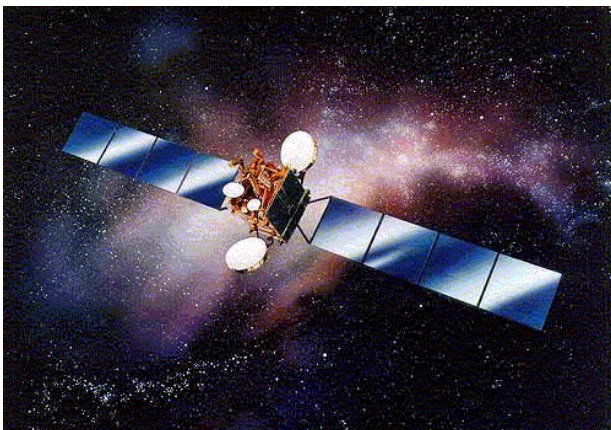


GSM 900 MHz
DCS 1800 MHz
UMTS 2 GHz



Wifi / Bluetooth
2.4 to 6 GHz

Anti-collision radar ~80 GHz
433 MHz remote opening



Satellite systems 1 to 45 GHz
(e.g. TV 12 GHz, GPS 1.5 GHz)



Military ship

Antennes à très larges bandes
du système de "guerre électronique" :
écoute des émissions adverses
(radars et communications)
pour identification et brouillage

Antennes de communication
(HF, VHF, etc.)

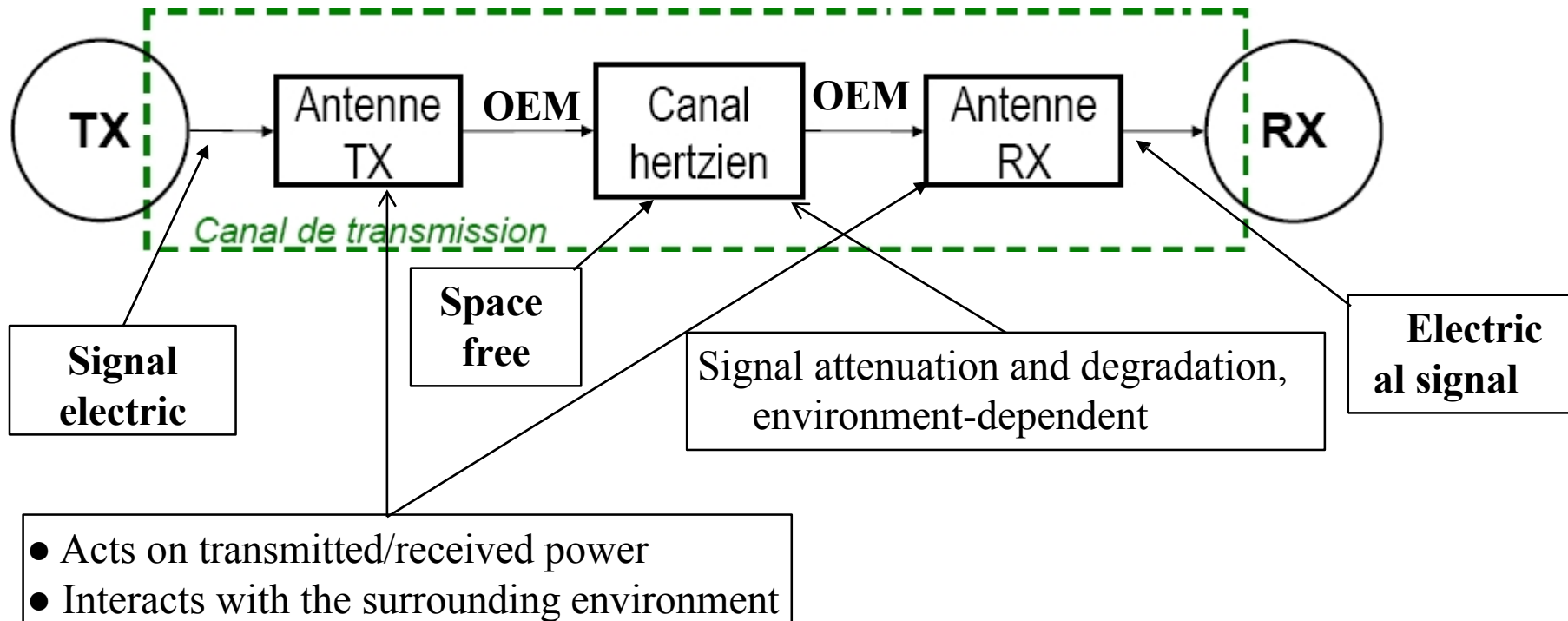
Antenne du radar
de conduite de tir
(artillerie)

Antenne des radars
d'acquisition et de poursuite
du système de lance-missiles

Antenne réseau (dalle)
du radar de veille aérienne
et de surface

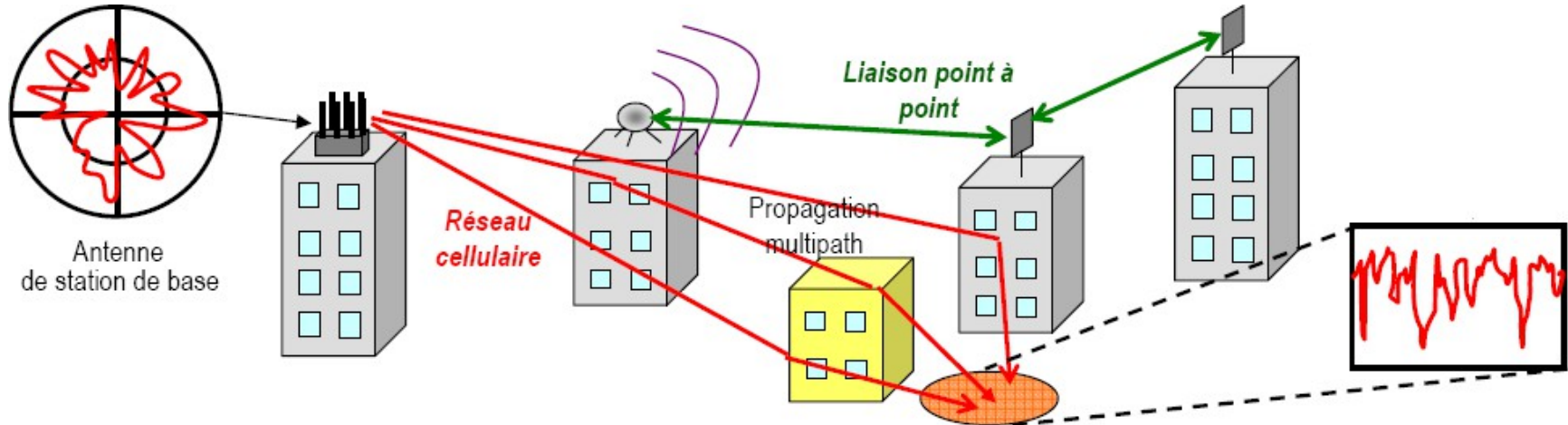


Transmission channel

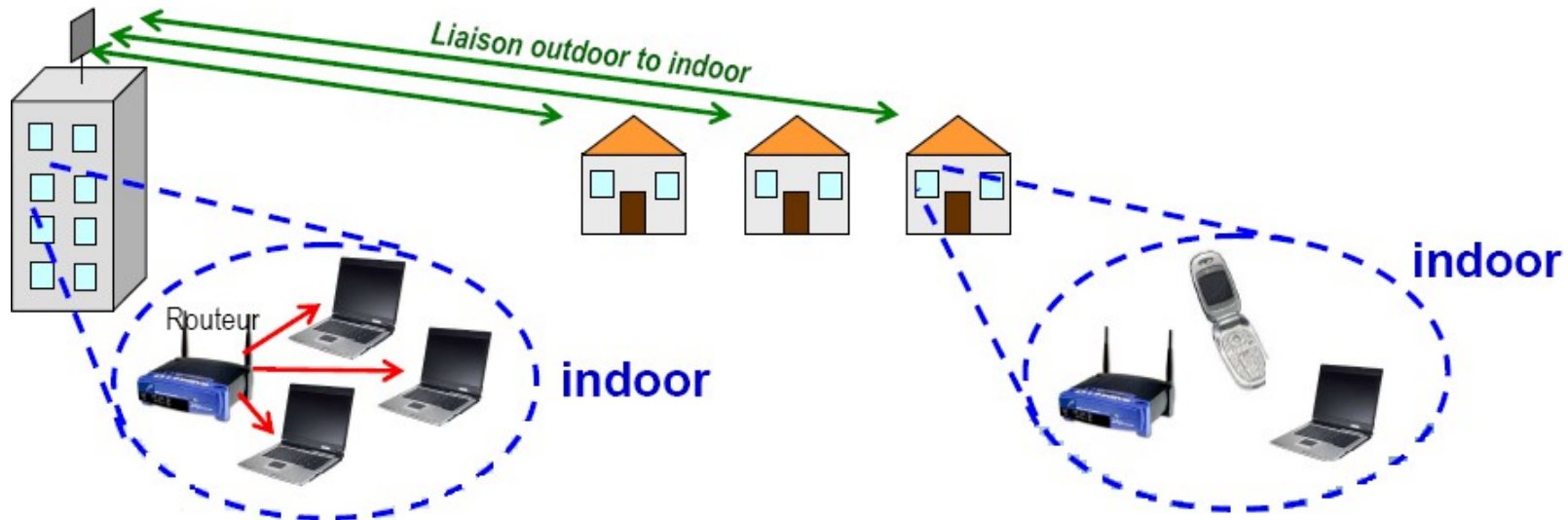


Channel for mobile telecommunications

Propagation in an urban environment



Propagation in an indoor environment



Areas covered

To understand the phenomena governing the operation of antennas and their integration into a communication system, the following **prerequisites must be met:**

- basic knowledge of electromagnetism (electromagnetic equations) Maxwell, EM wave propagation, etc.);
 - a good understanding of the principles of transmission lines and, above all, of the impedance matching of systems (travelling waves, standing waves, etc.);
- ★ antennas represent a practical application of Maxwell's equations where we can see physical phenomena without remaining purely in theoretical calculations.